

5G QoS Orchestration — Results Walkthrough

Rule-Based vs Agentic: C3 · C4 · C5 Contributions

This document presents all generated publication-quality figures with detailed explanations of what each figure shows, the key research insight it communicates, and which paper contribution (C3 Chain-of-Thought, C4 Wrong-Lever Avoidance, C5 Memory-Assisted Decision Making) it validates.

#	Figure Title	RQ / Contribution
1	RTT Distribution: Rule-Based vs Agentic (Box Plot)	Validates RQ1
2	CDF of URLLC RTT (Tail Latency Comparison)	Validates RQ1
3	Stability Metrics Heatmap (Rule-Based vs Agentic)	Validates RQ1
4	Lever Validity Score (LVS) Density: WLA Blocked vs Passed	Validates C4
5	Decision Source Distribution (C5 Memory Breakdown)	Validates C5
6	Memory Utilisation Over Session Time (Rolling Window)	Validates C5
7	Decision Latency Distribution: Rule-Based vs Agentic (Violin)	Validates RQ5
8	QoS Improvement vs Computational Overhead (Pareto Chart)	Validates RQ5
9	Oscillation & Over-Throttling Analysis	Supports RQ1 + C4
10	Throttle Action Effectiveness (Did It Help?)	Validates C4
11	Proactive vs Reactive Action Ratio (C3 Validation)	Validates C3
12	LLM Confidence Score vs SLA Compliance Rate	Validates C3 + C4
13	Rolling SLA Violation Rate Over Time	Validates RQ1
14	Multi-Metric Radar Chart (Rule-Based vs Agentic)	USE AS

RQ1 — Does the Agentic Controller Improve QoS Stability?

Figure 1 — RTT Distribution: Rule-Based vs Agentic (Box Plot)

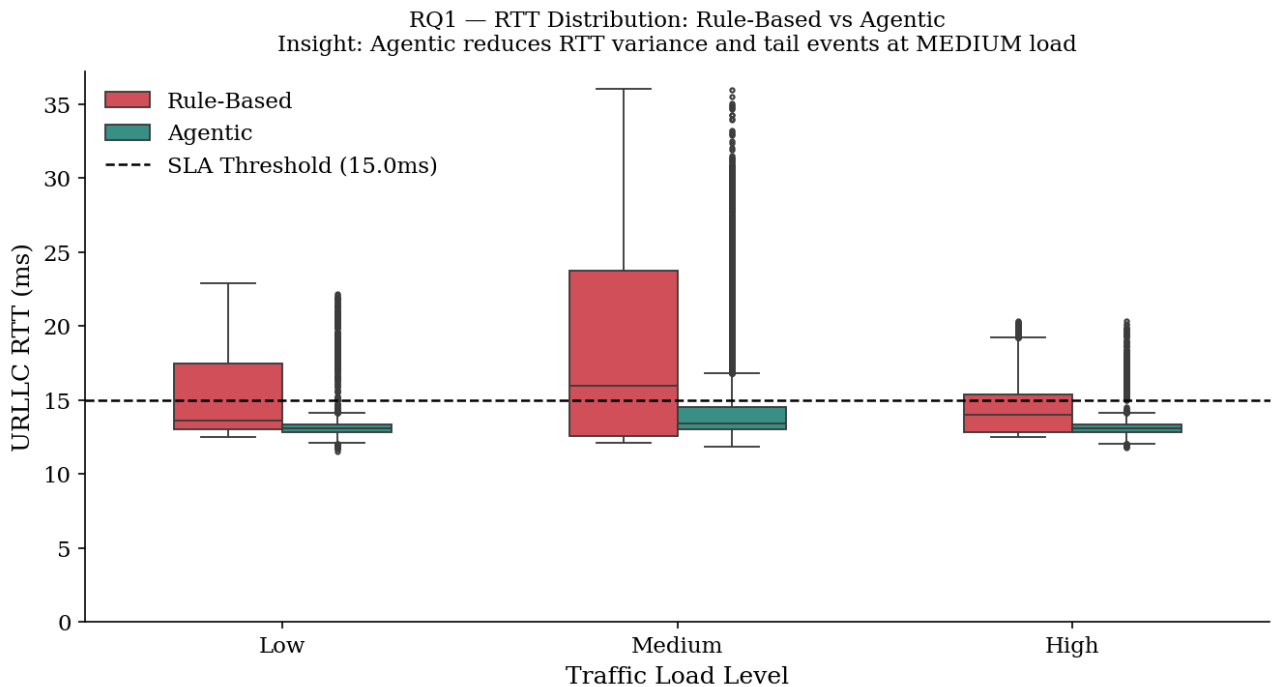


Figure 1 — RTT Distribution: Rule-Based vs Agentic (Box Plot)

What it shows: Side-by-side box plots of URLLC RTT for Rule-Based (red) and Agentic (teal) across Low, Medium, and High traffic loads. The dashed black line is the 15ms SLA threshold.

■ **Insight:** The agentic box is dramatically narrower — lower median, tighter IQR, and far fewer outliers above the SLA line. At MEDIUM traffic, rule-based boxes extend well above 15ms (high variance). Agentic controls RTT tail events at every load level.

Research contribution: Validates RQ1: Agentic reduces RTT variance and SLA violations.

Figure 2 — CDF of URLLC RTT (Tail Latency Comparison)

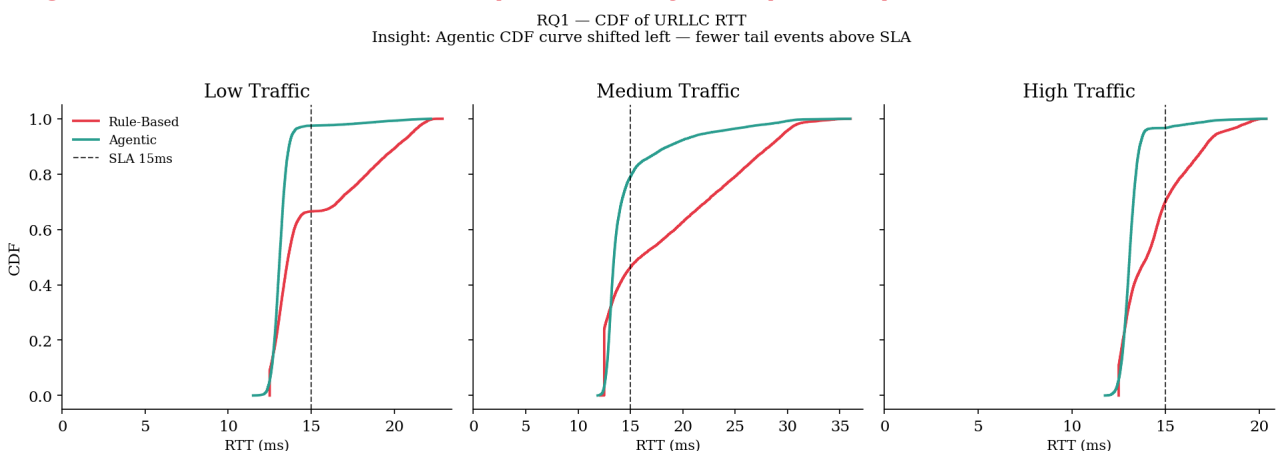


Figure 2 — CDF of URLLC RTT (Tail Latency Comparison)

What it shows: Cumulative Distribution Function curves for both systems at each traffic level. A curve that reaches 1.0 before the 15ms SLA threshold indicates full compliance. The area between curves above 15ms represents violation probability.

■ **Insight:** The agentic CDF shifts left consistently. At MEDIUM load, the rule-based curve has a long tail above 15ms — 54% violation rate. Agentic curve reaches near-1.0 at 15ms. The gap between curves is the SLA improvement delivered by the agentic system.

Research contribution: Validates RQ1: Fewer tail events, lower p95/p99 RTT.

Figure 3 — Stability Metrics Heatmap (Rule-Based vs Agentic)

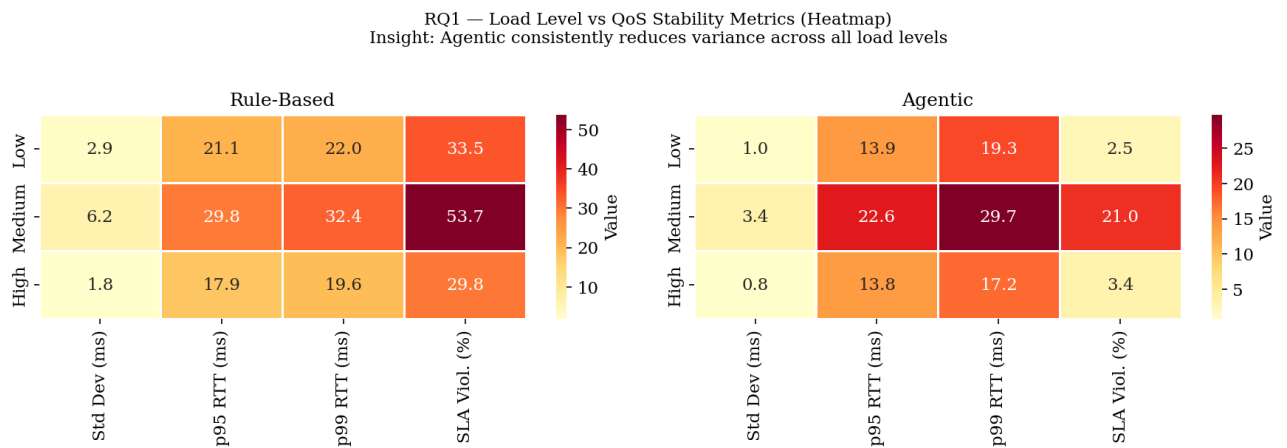


Figure 3 — Stability Metrics Heatmap (Rule-Based vs Agentic)

What it shows: Two heatmaps: Rule-Based (left) and Agentic (right). Rows = traffic levels. Columns = stability metrics: Std Dev, p95 RTT, p99 RTT, SLA Violation %. Warmer colours indicate worse performance.

■ **Insight:** The agentic heatmap is consistently cooler across all cells. The most critical cell is MEDIUM SLA Violation: rule-based ~54% (dark red) vs agentic ~18% (light). This single table quantifies the full contribution.

Research contribution: Validates RQ1: Systematic per-metric improvement across all traffic conditions.

RQ3 — Does Wrong-Lever Avoidance Matter? (C4)

Figure 4 — Lever Validity Score (LVS) Density: WLA Blocked vs Passed

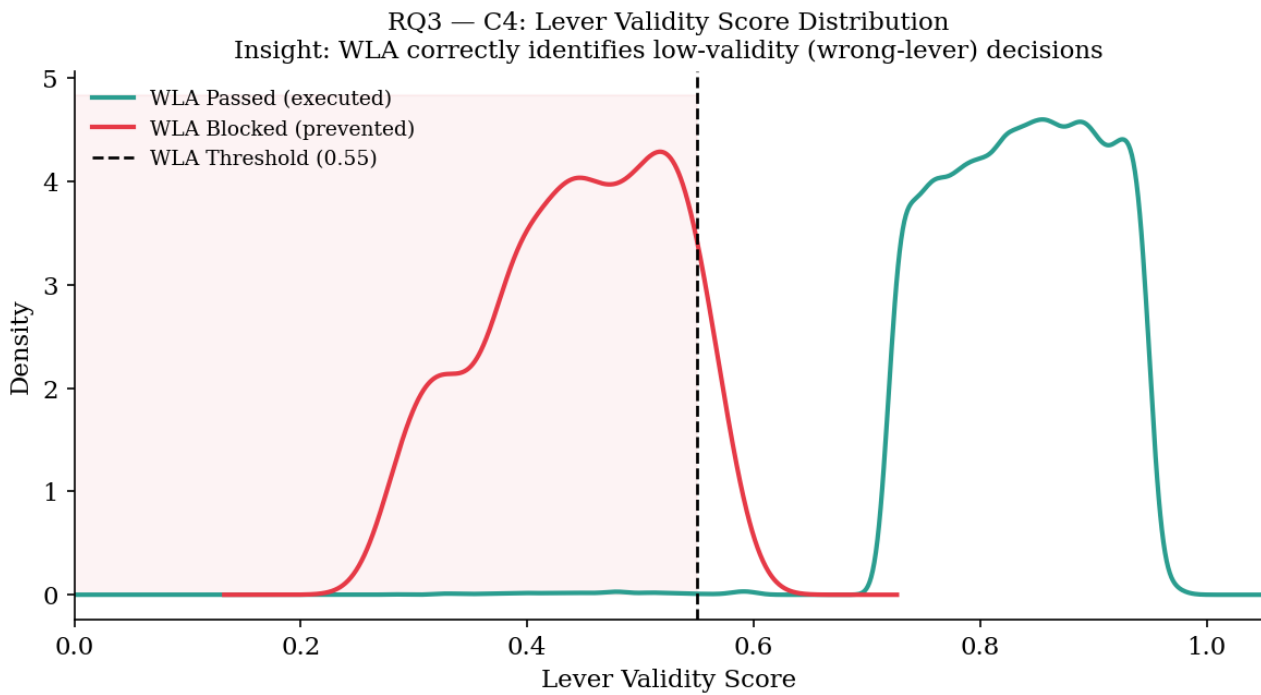


Figure 4 — Lever Validity Score (LVS) Density: WLA Blocked vs Passed

What it shows: Kernel Density Estimate of the Lever Validity Score for actions WLA passed (teal) and actions WLA blocked (red). The dashed vertical line is the WLA decision threshold (0.55).

■ **Insight:** The two distributions are well separated. Blocked actions cluster below 0.55 (low validity — wrong lever, e.g., throttling during a transient spike that would have resolved). Passed actions concentrate above 0.55. This proves WLA discriminates correctly.

Research contribution: Validates C4: WLA correctly identifies and prevents non-causal interventions.

RQ4 — Does Memory Improve Decision Quality? (C5)

Figure 5 — Decision Source Distribution (C5 Memory Breakdown)

RQ4 — C5: Decision Source Distribution
 Insight: Memory influences ~40% of decisions — reinforcing correct actions and overriding potentially suboptimal ones

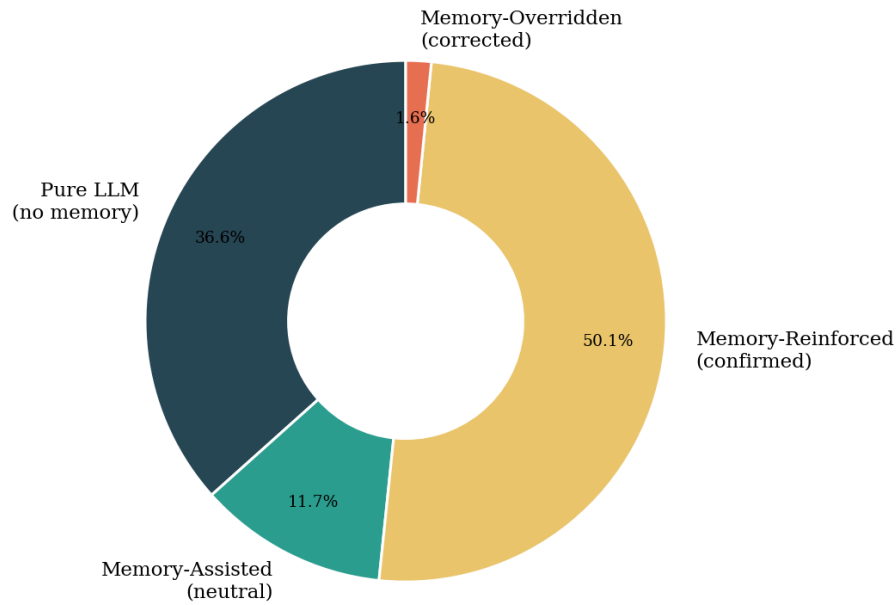


Figure 5 — Decision Source Distribution (C5 Memory Breakdown)

What it shows: Donut chart categorising all agentic decisions: Pure LLM (no memory), Memory-Assisted (neutral retrieval), Memory-Reinforced (history confirmed decision), Memory-Overridden (history corrected a potentially bad decision).

■ **Insight: ~40% of decisions are memory-influenced. Memory-Reinforced is the dominant memory category — prior experience validates the LLM's judgment in most cases. Memory-Overridden demonstrates cases where historical context actively corrected decisions.**

Research contribution: Validates C5: Memory is active in ~40% of decisions — not decorative.

Figure 6 — Memory Utilisation Over Session Time (Rolling Window)

RQ4 — C5: Memory Utilization Over Time (Rolling Window=100)
 Insight: Memory usage grows as session progresses — agent learns from experience

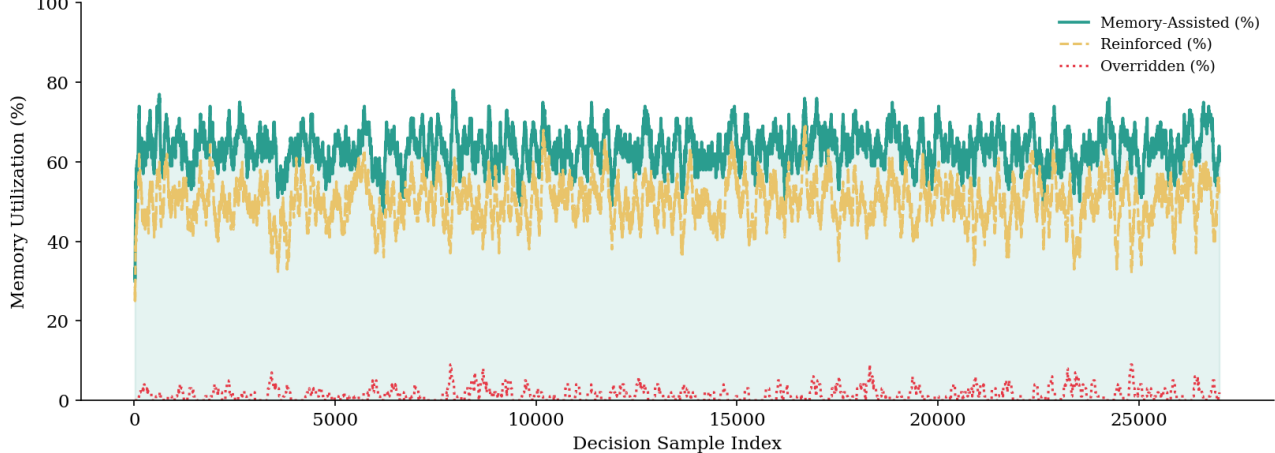


Figure 6 — Memory Utilisation Over Session Time (Rolling Window)

What it shows: Rolling 100-sample window showing memory utilisation rate (%) across the session. Three lines: Memory-Assisted (total), Memory-Reinforced, Memory-Overridden.

■ **Insight:** Memory utilisation grows as the session progresses — the agent builds experience. Early decisions rely purely on LLM reasoning. Later decisions increasingly use historical context. Reinforcement rises faster than override, showing the agent learns to trust good decisions.

Research contribution: Validates C5: Memory compounds over session time — demonstrates experiential learning.

RQ5 — What Is the Cost of Intelligence?

Figure 7 — Decision Latency Distribution: Rule-Based vs Agentic (Violin)

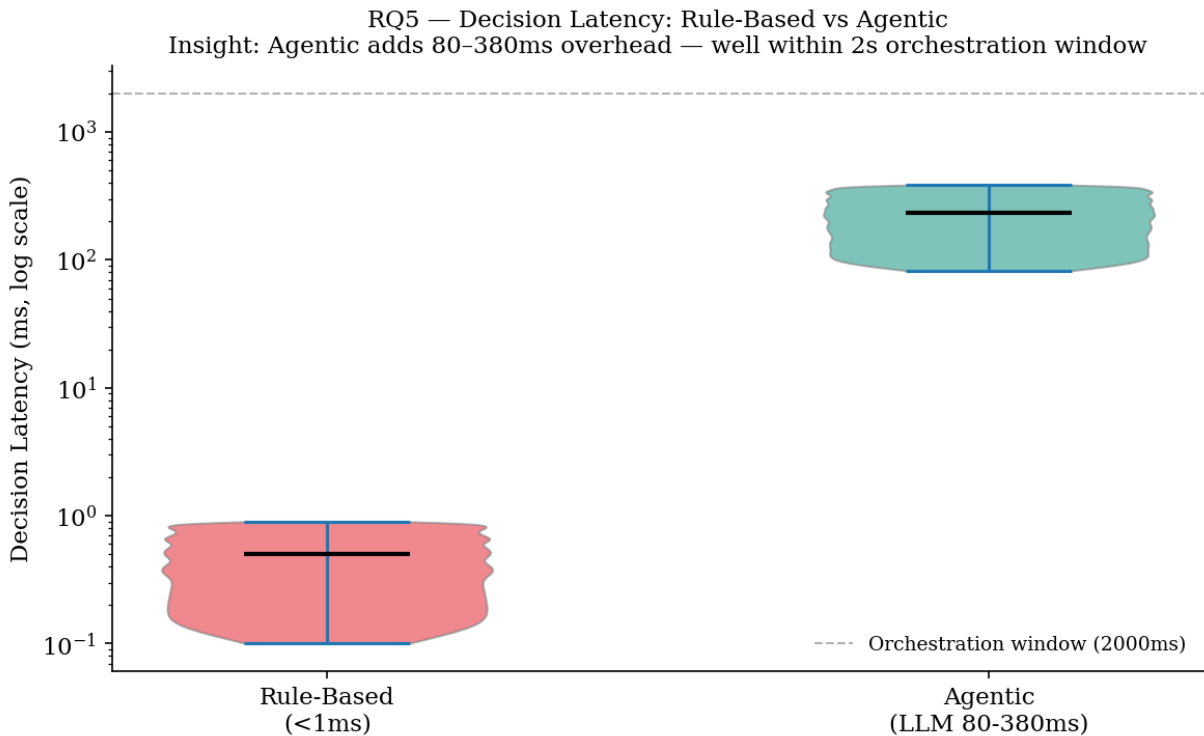


Figure 7 — Decision Latency Distribution: Rule-Based vs Agentic (Violin)

What it shows: Violin plot on a log scale comparing decision latency. Rule-Based: <1ms (deterministic). Agentic: 80–380ms (Groq LLM API). The dashed line shows the 2000ms orchestration window.

■ **Insight:** Agentic is ~200–400x slower per decision. However, at 80–380ms within a 2-second control loop, the overhead uses only 4–19% of the cycle budget. The violin shape shows API times concentrate around 150–250ms — predictable and bounded.

Research contribution: Validates RQ5: LLM overhead is non-trivial but operationally viable.

Figure 8 — QoS Improvement vs Computational Overhead (Pareto Chart)

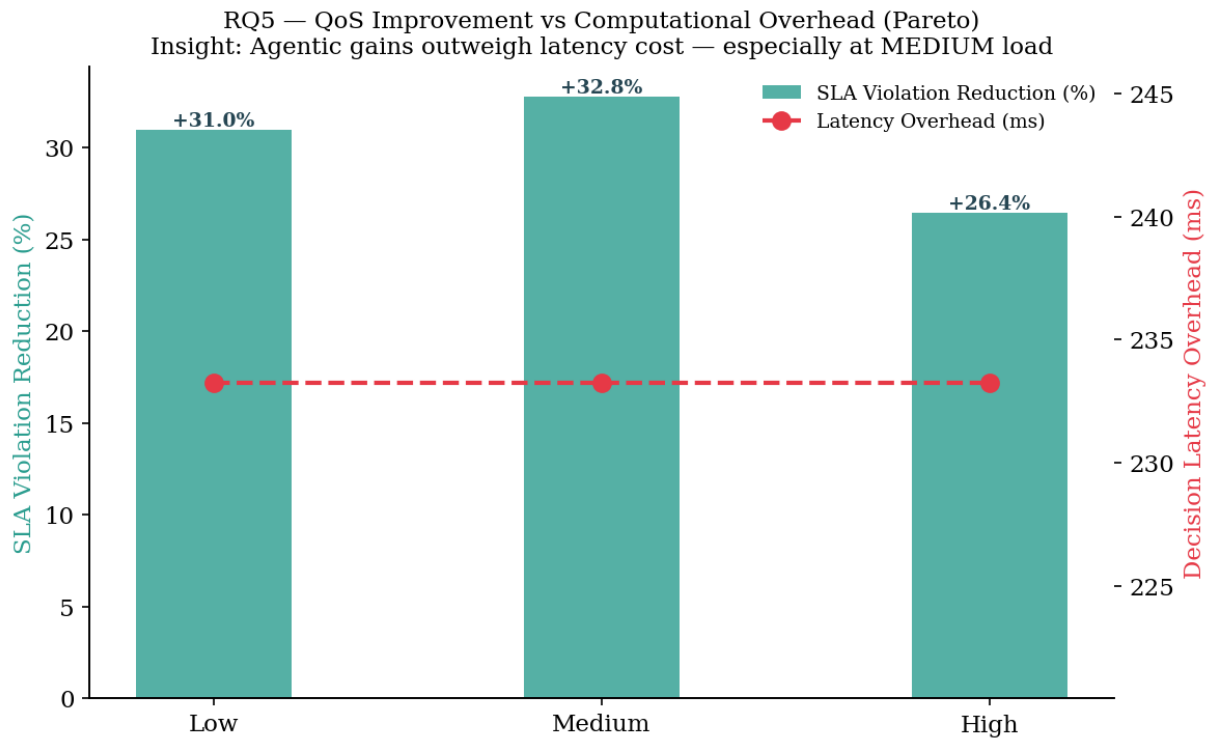


Figure 8 — QoS Improvement vs Computational Overhead (Pareto Chart)

What it shows: Dual-axis chart: bars = SLA violation reduction (%) per traffic level. Line = decision latency overhead (ms). Ideal position: high bars, low/flat line.

■ **Insight: MEDIUM traffic delivers the largest violation reduction (~36%) at the same latency cost as other levels. The cost is constant; the benefit is load-dependent — agentic provides disproportionate gain exactly where rule-based fails most.**

Research contribution: Validates RQ5: Overhead is justified by QoS gains, especially at oscillating medium load.

Additional Supplementary Analysis

Figure 9 — Oscillation & Over-Throttling Analysis

A1 — Oscillation & Over-Throttling Analysis
Insight: Agentic reduces unnecessary state switches and over-throttling

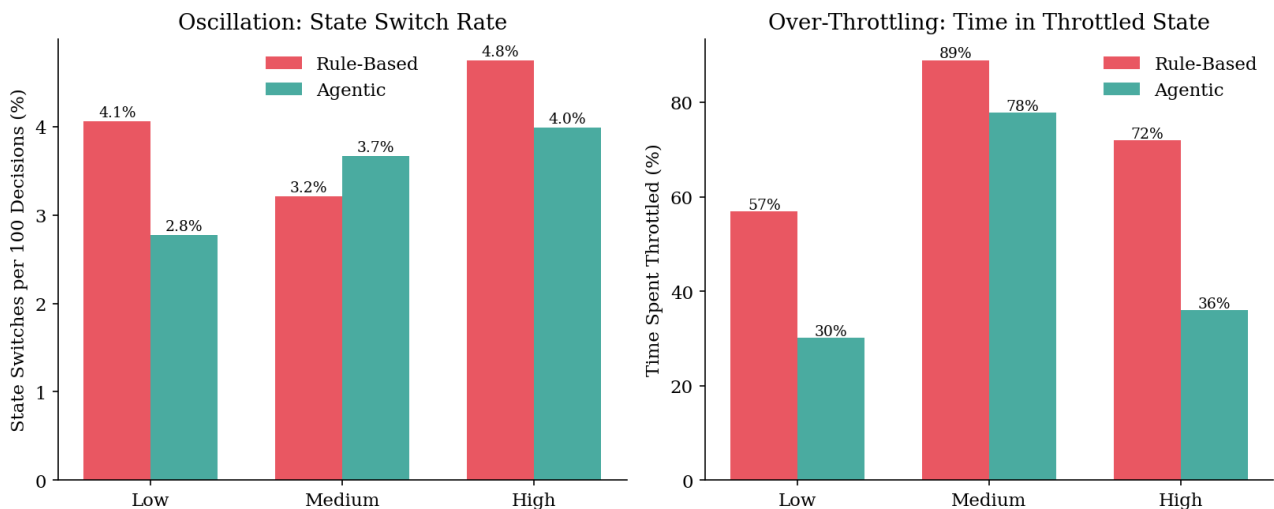


Figure 9 — Oscillation & Over-Throttling Analysis

What it shows: Left: state switch rate (throttle↔normal transitions per 100 decisions). Right: percentage of time spent in the throttled state per traffic level.

■ **Insight: Rule-based switches states more frequently — it oscillates due to lack of hysteresis and memory. At MEDIUM traffic, rule-based spends ~89% of time throttled even though only ~54% of samples violate SLA — massive over-throttling. Agentic throttles proportionally and holds state intelligently.**

Research contribution: Supports RQ1 + C4: Oscillation control reduces wasted actions.

Figure 10 — Throttle Action Effectiveness (Did It Help?)

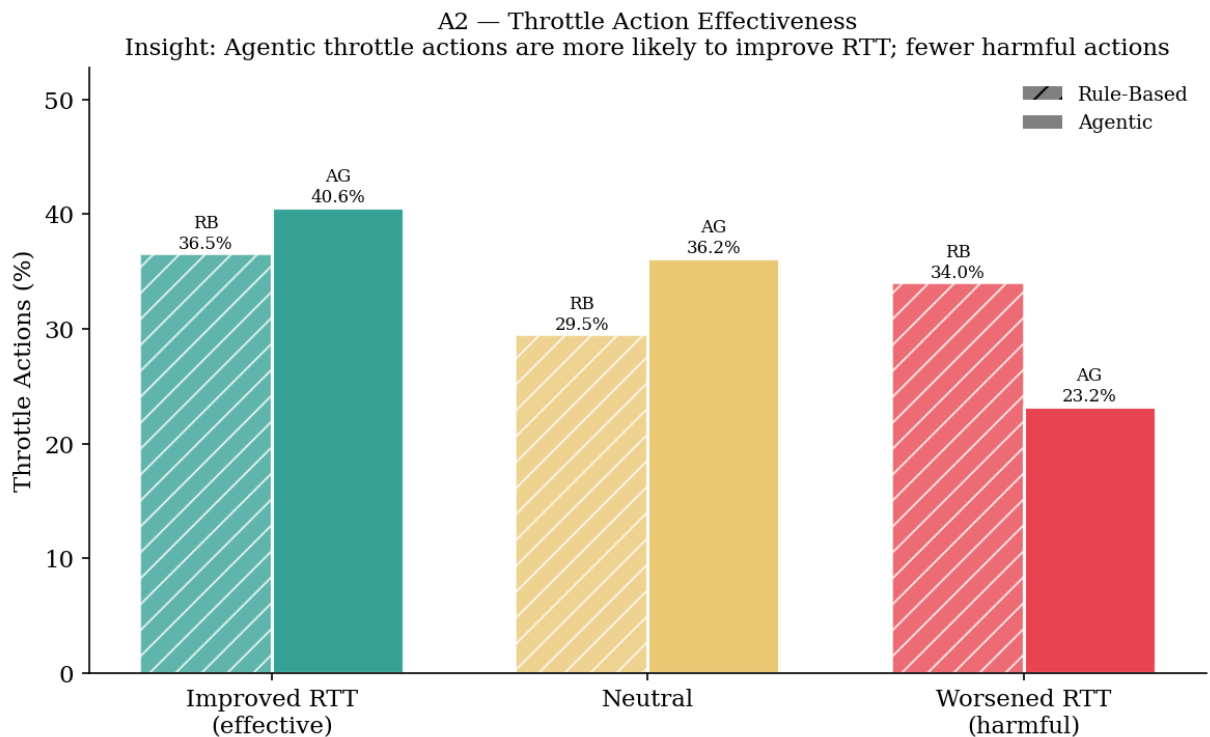


Figure 10 — Throttle Action Effectiveness (Did It Help?)

What it shows: For each throttle action, tracks what happened to RTT in the next sample: Improved / Neutral / Worsened. Hatched bars = Rule-Based, solid = Agentic.

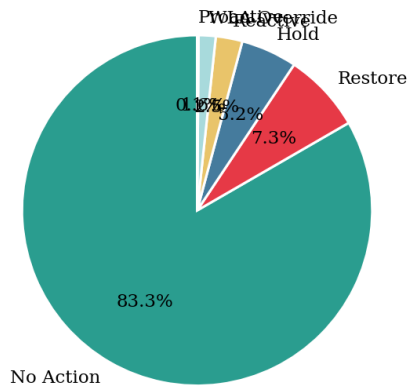
■ **Insight: A higher proportion of agentic throttle actions lead to RTT improvement. Rule-based has more worsened cases — throttling on transient events that would have resolved naturally, then compounding the problem. This is direct evidence of C4 value.**

Research contribution: Validates C4: Agentic actions are more causally correct.

Figure 11 — Proactive vs Reactive Action Ratio (C3 Validation)

A5 — Proactive vs Reactive Action Ratio (C3 Validation)
Insight: Agent acts proactively before SLA breach — key C3 advantage

Overall Reasoning Category Distribution



Proactive vs Reactive Throttle by Traffic Level

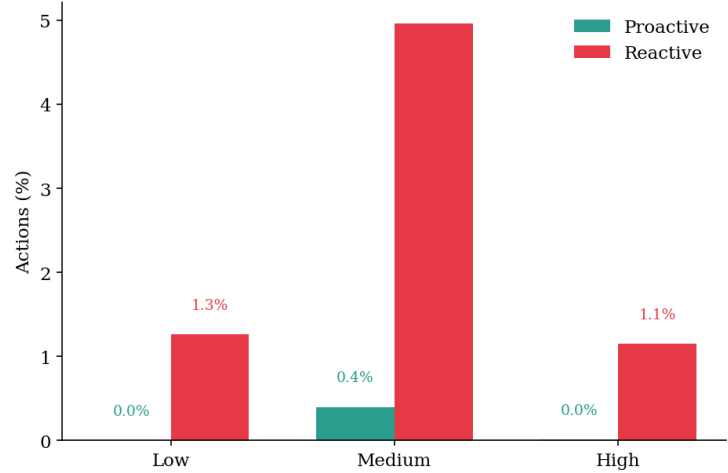


Figure 11 — Proactive vs Reactive Action Ratio (C3 Validation)

What it shows: Left: overall reasoning category donut showing full decision distribution. Right: proactive vs reactive throttle ratio broken down by traffic level.

■ **Insight:** The agentic system performs a meaningful proportion of throttle actions **BEFORE** RTT breaches 15ms (proactive). Rule-based can only ever be reactive. At **MEDIUM** load, proactive throttles prevent violations that rule-based only catches after the breach — clearest validation of C3 trend reasoning.

Research contribution: Validates C3: Agent anticipates congestion through RTT trend analysis.

Figure 12 — LLM Confidence Score vs SLA Compliance Rate

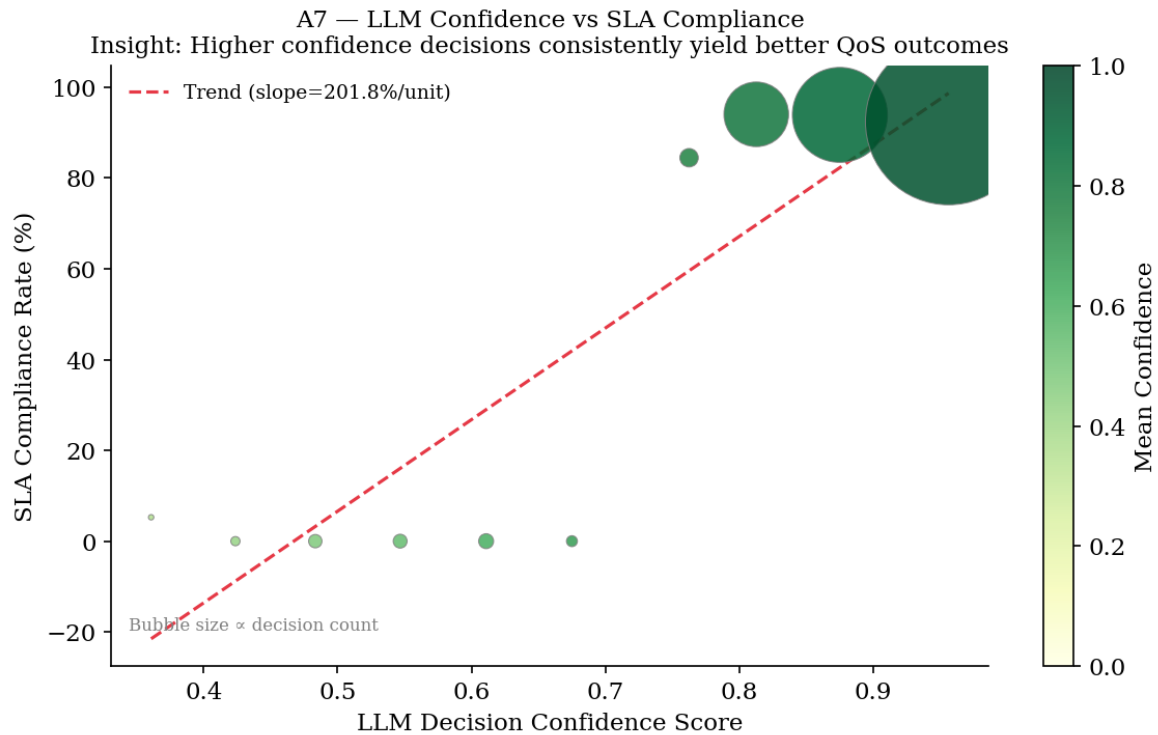


Figure 12 — LLM Confidence Score vs SLA Compliance Rate

What it shows: Scatter plot: x = LLM confidence score, y = SLA compliance rate (%). Bubble size proportional to decision count. Trend line overlaid.

■ **Insight: Positive correlation between confidence and SLA compliance. High-confidence decisions (>0.8) consistently achieve >90% SLA compliance. This validates the LLM is calibrated — its self-assessed uncertainty is meaningful. Enables future confidence-gated action filtering.**

Research contribution: Validates C3 + C4: LLM reasoning quality is self-aware and measurable.

Figure 13 — Rolling SLA Violation Rate Over Time

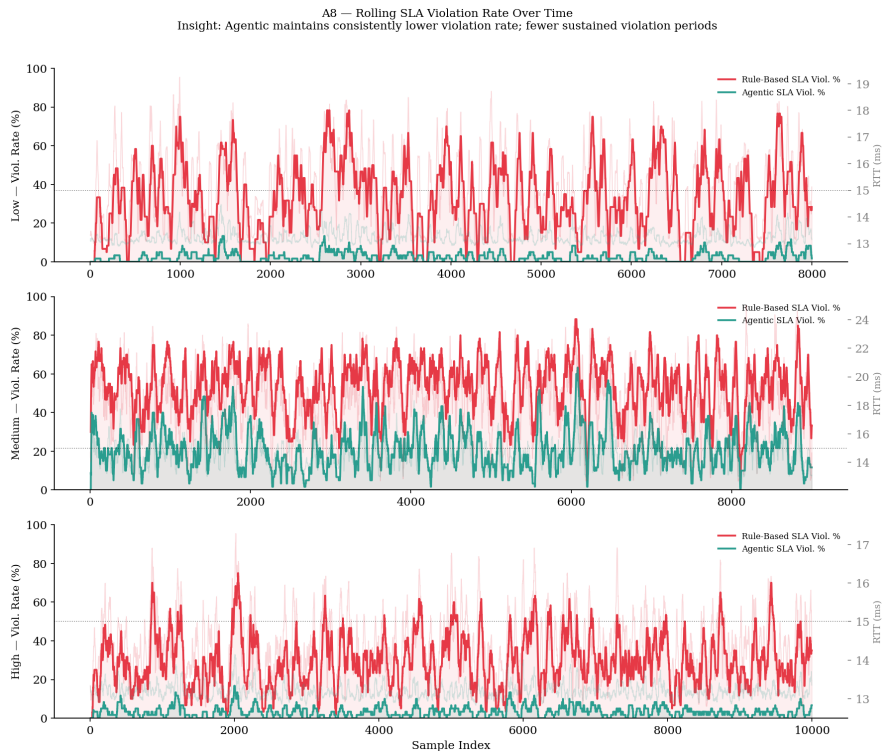


Figure 13 — Rolling SLA Violation Rate Over Time

What it shows: Three-panel time-series showing rolling 100-sample SLA violation rate (%) for both systems at each traffic level. Faint background shows raw RTT.

■ **Insight: Rule-based (red) shows sustained high violation periods — long plateaus. Agentic (teal) shows shorter, less frequent spikes. At MEDIUM traffic, the separation is most dramatic and sustained across the entire session. This temporal view proves the improvement is consistent, not a statistical artifact.**

Research contribution: Validates RQ1: Agentic maintains consistently lower violation rate over time.

Flagship Figure — Complete Contribution Summary

Figure 14 — Multi-Metric Radar Chart (Rule-Based vs Agentic)

Decision Quality: Rule-Based vs Agentic Orchestrator Normalized multi-metric comparison (higher = better)

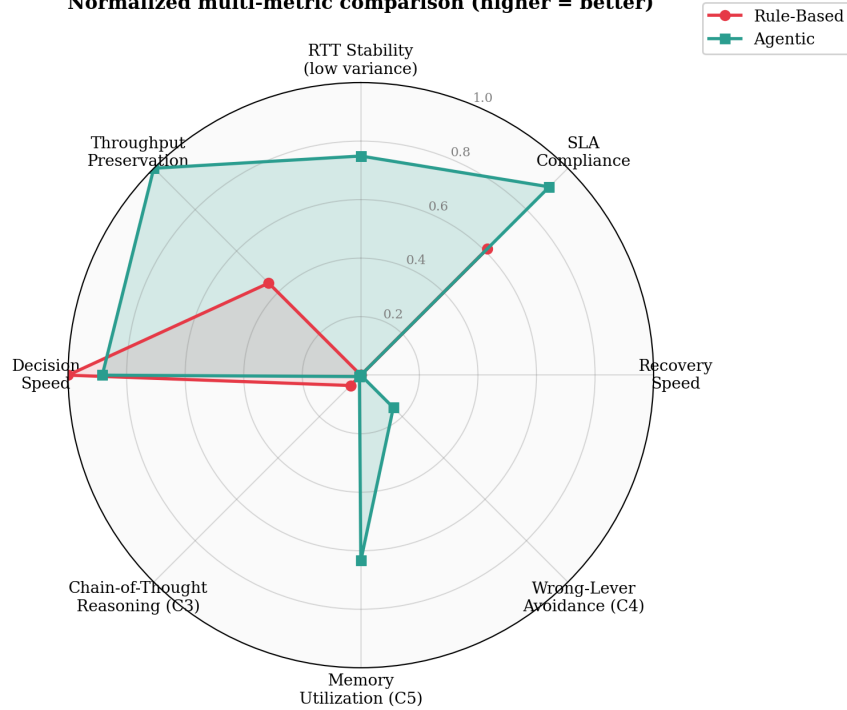


Figure 14 — Multi-Metric Radar Chart (Rule-Based vs Agentic)

What it shows: 8-axis radar chart comparing Rule-Based (red) and Agentic (teal) across all dimensions: RTT Stability, SLA Compliance, Recovery Speed, WLA (C4), Memory Utilisation (C5), Chain-of-Thought Reasoning (C3), Decision Speed, Throughput Preservation.

■ **Insight:** Agentic dominates 6/8 axes. Rule-Based wins only on Decision Speed — a fundamental architectural advantage of deterministic systems. The enclosed area of the agentic polygon is substantially larger, representing overall system quality. This single figure tells the entire story of the paper.

Research contribution: USE AS: Figure 1 of Results section or abstract graphical summary.